

Theorem F — the coset construction of the Klein map

PLANETVORTEX — the point monograph

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Statement

Let $G = PSL(2, 7)$ (168 classes over F_7) and let (a, b) be any pair with $a^2 = b^3 = (ab)^7 = 1$. With $X = \langle a \rangle$, $Y = \langle b \rangle$, $Z = \langle ab \rangle$, define the vertices as the right cosets gY , the edges as gX , the faces as gZ , with the incidence by nonempty coset intersection. Then there are 56 vertices, 84 edges, 24 faces and $V - E + F = -4$; every vertex has degree 3, every face is a 7-cycle; the map is connected and orientable; the left action of G is faithful — the Klein map $\{7, 3\}_8$ with $Aut^{+ \circ} \cong PSL(2, 7)^\circ$ and $|Aut| = 336$ including reflections.

Proof

The number of right cosets of H is $|G|/|H|$: $168/3=56$, $168/2=84$, $168/7=24$. The stabilizer chain of the triangle group gives exactly 3 edges per vertex, 2 vertices and 2 faces per edge, and a 7-cycle of edges per face. The $(2,3,7)$ triple generates G , so the flag graph is connected; the full flag set of 336 pairwise-incident triples has a bipartite adjacency graph (two flags adjacent iff they differ in one member) — orientability. Hence $\chi = V - E + F = -4$, genus $g = 1 - \chi/2 = 3$. Faithfulness: the core of $\langle b \rangle$ is trivial in the simple group G . The Hurwitz bound $84(g-1) = 168$ caps the orientation-preserving automorphisms at the certified 168; the reflections double the count. $\square$

Computational certificate

Stage V2 (V2_klein_tiling_default.json): the exact combinatorial registers, the connected and bipartite flag graph, and the $PGL(2, 7)$ flag certificate on all 336×3 flag moves.